

Praktikum 1 - Struktur Sistem Operasi



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Praktik 1.1: Menjelajahi Layanan OS

```
yoga@MSI: /tmp/test_dir

Codename:       noble
yoga@MSI:~$ ps aux | head -20
USER        PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.2  0.3 21784 12252 ?        Ss   14:51   0:01 /sbin/init
root         2  0.0  0.0   3072 1792 ?        Sl   14:51   0:00 /init
root         7  0.0  0.0   3072 1792 ?        Sl   14:51   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        43  0.1  0.3 66832 15328 ?        Ss   14:51   0:00 /usr/lib/systemd/systemd-journald
root        93  0.2  0.1 25120 6348 ?        Ss   14:51   0:00 /usr/lib/systemd/systemd-udevd
systemd+   108  0.0  0.3 21456 12672 ?        Ss   14:51   0:00 /usr/lib/systemd/systemd-resolved
root       109  0.0  0.1 24704 4420 ?        S   14:51   0:00 (udev-worker)
root       110  0.0  0.1 24704 4152 ?        S   14:51   0:00 (udev-worker)
root       111  0.0  0.1 25172 4920 ?        S   14:51   0:00 (udev-worker)
root       112  0.0  0.1 24704 4028 ?        S   14:51   0:00 (udev-worker)
root       113  0.0  0.1 24704 4292 ?        S   14:51   0:00 (udev-worker)
root       114  0.0  0.1 24704 3988 ?        S   14:51   0:00 (udev-worker)
root       115  0.0  0.1 24704 4192 ?        S   14:51   0:00 (udev-worker)
root       116  0.1  0.1 25172 5192 ?        S   14:51   0:00 (udev-worker)
root       117  0.0  0.1 24704 4048 ?        S   14:51   0:00 (udev-worker)
systemd+   118  0.0  0.1 91824 7680 ?        Ssl  14:51   0:00 /usr/lib/systemd/systemd-timesyncd
root       119  0.0  0.1 24704 4340 ?        S   14:51   0:00 (udev-worker)
root       120  0.0  0.1 24980 4492 ?        S   14:51   0:00 (udev-worker)
root       122  0.0  0.1 24704 3984 ?        S   14:51   0:00 (udev-worker)

yoga@MSI:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.7Gi        433Mi       2.9Gi         3.5Mi        457Mi       3.3Gi
Swap:           1.0Gi          0B         1.0Gi

yoga@MSI:~$ lscpu
Architecture:        x86_64
CPU op-mode(s):      32-bit, 64-bit
Address sizes:        39 bits physical, 48 bits virtual
Byte Order:           Little Endian
CPU(s):               16
On-line CPU(s) list: 0-15
Vendor ID:            GenuineIntel
Model name:           12th Gen Intel(R) Core(TM) i7-12650H
CPU family:           6
Model:                154
Thread(s) per core:   2
Core(s) per socket:   8
Socket(s):            1
Stepping:             1
BogoMIPS:             5375.99
Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr p
ge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss
ht syscall nx pdpeb3 vtscpl lm constant_tsc rep
good_nopl xtopology tsc_reliable nonstop_tsc cpuid
tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 p
cid sse4_1 sse4_2 v2apic movbe popcnt tsc_deadline
_timer aes xsave avx f16c rdrand hypervisor lahf_l
m abm 3dnowprefetch ssbd ibrs ibpb stibp ibrs_enha
nced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjus
t bti avx2 smep bmi2 erms invpcid rdseed adx saop
clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xs
aves avx_vnni vmmi umip waitpkg gfni vaes vpclmulq
dq rdpid movdiri movdir64b fsrm md_clear serialize
flush_l3d arch_capabilities

Virtualization features:
Virtualization:       VT-x
Hypervisor vendor:    Microsoft
Virtualization type:   full
Caches (sum of all):
  L1d:                 384 KiB (8 instances)
  L1i:                 256 KiB (8 instances)
  L2:                  10 MiB (8 instances)
  L3:                  24 MiB (1 instance)
NUMA:
NUMA node(s):         1
NUMA node0 CPU(s):    0-15
Vulnerabilities:
Gather data sampling: Not affected
Itlb multihit:        Not affected
L1tf:                 Not affected
Mds:                  Not affected
Meltdown:             Not affected
Mmio stale data:      Not affected
Reg file data sampling: Vulnerable: No microcode
Retbleed:             Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass:    Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:           Mitigation; usercopy/swapgs barriers and __user po
inter sanitization
Spectre v2:           Mitigation; Enhanced / Automatic IBRS; IBPB condit
ional; RSB filling; PBSRB-eIBRS SW sequence; BMI B
MI_OIS_5
Srbds:                Not affected
Tsx async abort:      Not affected

yoga@MSI:~$ lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda  8:0  0 388.4M 1 disk
sdb  8:16 0 186M 1 disk
sdc  8:32 0 1G 0 disk [swap]
sdd  8:48 0 1T 0 disk /mnt/wslg/distro
sde  8:64 0 1T 0 disk
```

Praktik 1.2: Program C - Menggunakan Layanan OS

```
yoga@MSI:~$ nano os_services.c
yoga@MSI:~$ gcc -o os_services os_services.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:~$ ./os_services
-bash: ./os_services: No such file or directory
```

2. User and Operating-System Interface

Praktik 2.1: Command Line Interface (CLI)

Praktik 2.2: Program C - Shell Sederhana Buat file simple_shell.c:

```
yoga@MSI:~$ pwd
/home/yoga
yoga@MSI:~$ ls -la
total 64
drwxr-xr-x 4 yoga yoga 4096 Oct 16 14:58 .
drwxr-xr-x 3 root root 4096 Oct 16 02:32 ..
-rw-r--r-- 1 yoga yoga 691 Oct 16 03:49 .bash_history
-rw-r--r-- 1 yoga yoga 220 Oct 16 02:32 .bash_logout
-rw-r--r-- 1 yoga yoga 3771 Oct 16 02:32 .bashrc
drwxr-xr-x 2 yoga yoga 4096 Oct 16 02:32 .cache
-rw-r--r-- 1 yoga yoga 20 Oct 16 03:09 .lessht
drwxr-xr-x 3 yoga yoga 4096 Oct 16 02:37 .local
-rw-rw-r-- 1 yoga yoga 0 Oct 16 02:32 .motd_shown
-rw-r--r-- 1 yoga yoga 807 Oct 16 02:32 .profile
-rw-r--r-- 1 yoga yoga 1465 Oct 16 03:03 e
-rw-r--r-- 1 yoga yoga 1465 Oct 16 03:03 file_syscalls.c
-rw-r--r-- 1 yoga yoga 1078 Oct 16 03:06 info_syscalls.c
-rw-r--r-- 1 yoga yoga 1822 Oct 16 02:39 os_services.c
-rw-r--r-- 1 yoga yoga 12 Oct 16 02:47 output.txt
-rw-r--r-- 1 yoga yoga 1082 Oct 16 02:58 process_syscalls.c
-rw-r--r-- 1 yoga yoga 1349 Oct 16 02:53 simple_shell.c
yoga@MSI:~$ cd /tmp
yoga@MSI:/tmp$ mkdir test_dir
yoga@MSI:/tmp$ cd test_dir
yoga@MSI:/tmp/test_dir$ ls -l | grep "^d"
yoga@MSI:/tmp/test_dir$ ps aux | grep bash | wc -l
3
yoga@MSI:/tmp/test_dir$ echo "Hello World" > output.txt
yoga@MSI:/tmp/test_dir$ cat output.txt
Hello World
yoga@MSI:/tmp/test_dir$ cat output.txt >> output.txt
cat: output.txt: input file is output file
yoga@MSI:/tmp/test_dir$ cat < output.txt
Hello World
yoga@MSI:/tmp/test_dir$ sleep 30 &
[1] 520
yoga@MSI:/tmp/test_dir$ jobs
[1]+  Running                  sleep 30 &
yoga@MSI:/tmp/test_dir$ fg %1
sleep 30
```

```
yoga@MSI:/tmp/test_dir$ nano simple_shell.c
yoga@MSI:/tmp/test_dir$ gcc -o simple_shell simple_shell.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ ./simple_shell
-bash: ./simple_shell: No such file or directory
```

3. System Calls

Praktik 3.1: Process Control System Calls

Praktik 3.2: File Management System Calls

Praktik 3.2: File Management System Calls

```
yoga@MSI:/tmp/test_dir$ nano process_syscalls.c:
yoga@MSI:/tmp/test_dir$ nano process_syscalls.c
yoga@MSI:/tmp/test_dir$ nano file_syscalls.c
yoga@MSI:/tmp/test_dir$ nano e info_syscalls.c
yoga@MSI:/tmp/test_dir$ nano info_syscalls.c
yoga@MSI:/tmp/test_dir$ gcc -o process_syscalls process_syscalls.c
\Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc -o file_syscalls file_syscalls.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc -o info_syscalls info_syscalls.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ ./process_syscalls
-bash: ./process_syscalls: No such file or directory
yoga@MSI:/tmp/test_dir$ ./file_syscalls
-bash: ./file_syscalls: No such file or directory
yoga@MSI:/tmp/test_dir$ ./info_syscalls
-bash: ./info_syscalls: No such file or directory
yoga@MSI:/tmp/test_dir$ ^C
yoga@MSI:/tmp/test_dir$ systemctl list-units --type=service
```

4. System Services

Praktik 4.1: Menggunakan systemd

Praktik 4.2: Membuat Custom System Service

```
yoga@MSI:/tmp/test_dir$ journalctl -u ssh -n 20
-- No entries --
yoga@MSI:/tmp/test_dir$ systemctl list-unit-files --type=service --state=enabled
UNIT FILE                                STATE    PRESET
apparmor.service                        enabled  enabled
apport.service                          enabled  enabled
cloud-config.service                    enabled  enabled
cloud-final.service                     enabled  enabled
cloud-init-local.service                 enabled  enabled
cloud-init.service                      enabled  enabled
console-setup.service                   enabled  enabled
cron.service                            enabled  enabled
dmesg.service                           enabled  enabled
e2scrub_reap.service                    enabled  enabled
getty@.service                           enabled  enabled
keyboard-setup.service                  enabled  enabled
landscape-client.service                 enabled  enabled
networkd-dispatcher.service             enabled  enabled
rsyslog.service                         enabled  enabled
setvtrgb.service                        enabled  enabled
snapd.apparmor.service                  enabled  enabled
snapd.autoimport.service                enabled  enabled
snapd.core-fixup.service                enabled  enabled
snapd.recovery-chooser-trigger.service  enabled  enabled
snapd.seeded.service                    enabled  enabled
snapd.service                           enabled  enabled
snapd.system-shutdown.service           enabled  enabled
systemd-pstore.service                  enabled  enabled
systemd-resolved.service                 enabled  enabled
systemd-timesyncd.service                enabled  enabled
ua-reboot-cmds.service                  enabled  enabled
ubuntu-advantage.service                enabled  enabled
unattended-upgrades.service              enabled  enabled
wsl-pro.service                         enabled  enabled

30 unit files listed.
yoga@MSI:/tmp/test_dir$ nano my_daemon.c
yoga@MSI:/tmp/test_dir$ return 0;
}
-bash: return: can only 'return' from a function or sourced script
-bash: syntax error near unexpected token `}'
yoga@MSI:/tmp/test_dir$ nano gcc -o my_daemon my_daemon.c
Invalid operating directory: my_daemon
yoga@MSI:/tmp/test_dir$ gcc -o my_daemon my_daemon.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ nano /etc/systemd/system/my_daemon.service
yoga@MSI:/tmp/test_dir$ /etc/systemd/system/my_daemon.service:
-bash: /etc/systemd/system/my_daemon.service: No such file or directory
yoga@MSI:/tmp/test_dir$ nano sudo /etc/systemd/system/my_daemon.service:
yoga@MSI:/tmp/test_dir$ sudo nano /etc/systemd/system/my_daemon.service
[sudo] password for yoga:
yoga@MSI:/tmp/test_dir$ sudo systemctl daemon-reload
yoga@MSI:/tmp/test_dir$ sudo systemctl start my_daemon
```

5. Linkers and Loaders

Praktik 5.1: Proses Linking

Praktik 5.2: Static vs Dynamic Linking

Praktik 5.3: Membuat dan Menggunakan Shared Library

```
yoga@MSI:/tmp/test_dir$ nano /etc/systemd/system/my_daemon.service
yoga@MSI:/tmp/test_dir$ nano math_ops.h
yoga@MSI:/tmp/test_dir$ nano math_ops.c
yoga@MSI:/tmp/test_dir$ nano main.c
yoga@MSI:/tmp/test_dir$ gcc -E main.c -o main.i
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc -E math_ops.c -o math_ops.i
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc -S main.c -o main.s
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc -S math_ops.c -o math_ops.s
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc -c main.c -o main.o
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc -c math_ops.c -o math_ops.o
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc main.o math_ops.o -o program
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ nm main.o
nm: 'main.o': No such file
yoga@MSI:/tmp/test_dir$ nm math_ops.o
nm: 'math_ops.o': No such file
yoga@MSI:/tmp/test_dir$ nm program
nm: 'program': No such file
yoga@MSI:/tmp/test_dir$ ldd program
ldd: ./program: No such file or directory
yoga@MSI:/tmp/test_dir$ gcc main.c math_ops.c -o program_static -static
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc main.c math_ops.c -o program_dynamic
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ ls -lh program_static program_dynamic
ls: cannot access 'program_static': No such file or directory
ls: cannot access 'program_dynamic': No such file or directory
yoga@MSI:/tmp/test_dir$ ldd program_static
ldd: ./program_static: No such file or directory
yoga@MSI:/tmp/test_dir$ ldd program_dynamic
ldd: ./program_dynamic: No such file or directory
yoga@MSI:/tmp/test_dir$ gcc -c -fPIC math_ops.c -o math_ops.o
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc -shared -o libmathops.so math_ops.o
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gcc main.c -L. -lmathops -o program_shared
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ LD_LIBRARY_PATH=. ./program_shared
-bash: ./program_shared: No such file or directory
yoga@MSI:/tmp/test_dir$ readelf -d program_shared
```


6. Why Applications Are Operating-System Specific

Praktik 6.1: System Call Differences

Praktik 6.2: Binary Format Analysis

```
yoga@MSI:/tmp/test_dir$ nano os_specific.c
yoga@MSI:/tmp/test_dir$ nano os_specific.c
yoga@MSI:/tmp/test_dir$ file program
program: cannot open 'program' (No such file or directory)
yoga@MSI:/tmp/test_dir$ readelf -h program
readelf: Error: 'program': No such file
yoga@MSI:/tmp/test_dir$ readelf -S program
readelf: Error: 'program': No such file
yoga@MSI:/tmp/test_dir$ readelf -l program
readelf: Error: 'program': No such file
yoga@MSI:/tmp/test_dir$ objdump -d program | head -50
objdump: 'program': No such file
```


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Praktik 7.1: Kernel Module Sederhana

Praktik 7.2: Monitoring Kernel

```
yoga@MSI:/tmp/test_dir$ nano hello_module.c
yoga@MSI:/tmp/test_dir$ nano hello_module.c
yoga@MSI:/tmp/test_dir$ obj-m += hello_module.o
obj-m: command not found
yoga@MSI:/tmp/test_dir$ obj-m += hello_module.o
obj-m: command not found
yoga@MSI:/tmp/test_dir$ obj-m += hello_module.o
all:

yoga@MSI:/tmp/test_dir$ make
Command 'make' not found, but can be installed with:
sudo apt install make # version 4.3-4.1build1, or
sudo apt install make-guile # version 4.3-4.1build1
yoga@MSI:/tmp/test_dir$ sudo insmod hello_module.ko
[sudo] password for yoga:
insmod: ERROR: could not load module hello_module.ko: No such file or directory
yoga@MSI:/tmp/test_dir$ lsmod | grep hello
yoga@MSI:/tmp/test_dir$ dmesg | tail
[ 522.190893] systemd-journald[43]: Time jumped backwards, rotating.
[ 567.953437] mini_init (246): drop_caches: 1
[ 750.033321] systemd-journald[43]: Time jumped backwards, rotating.
[ 1877.985168] systemd-journald[43]: Time jumped backwards, rotating.
[ 2760.752533] systemd-journald[43]: Time jumped backwards, rotating.
[ 2897.305132] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 3004.266273] systemd-journald[43]: Time jumped backwards, rotating.
[ 3047.886305] mini_init (246): drop_caches: 1
[ 4128.464089] systemd-journald[43]: Time jumped backwards, rotating.
[ 4522.541685] systemd-journald[43]: Time jumped backwards, rotating.
yoga@MSI:/tmp/test_dir$ sudo rmmod hello_module
rmmod: ERROR: Module hello_module is not currently loaded
yoga@MSI:/tmp/test_dir$ dmesg | tail
[ 567.953437] mini_init (246): drop_caches: 1
[ 750.033321] systemd-journald[43]: Time jumped backwards, rotating.
[ 1877.985168] systemd-journald[43]: Time jumped backwards, rotating.
[ 2760.752533] systemd-journald[43]: Time jumped backwards, rotating.
[ 2897.305132] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 3004.266273] systemd-journald[43]: Time jumped backwards, rotating.
[ 3047.886305] mini_init (246): drop_caches: 1
[ 4128.464089] systemd-journald[43]: Time jumped backwards, rotating.
[ 4522.541685] systemd-journald[43]: Time jumped backwards, rotating.
[ 4748.311996] systemd-journald[43]: Time jumped backwards, rotating.
yoga@MSI:/tmp/test_dir$ uname -r
6.6.87.2-microsoft-standard-WSL2
yoga@MSI:/tmp/test_dir$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025
yoga@MSI:/tmp/test_dir$ sysctl -a | head -20
sysctl: permission denied on key 'kernel.cad_pid'
abi.vsyscall32 = 1
debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autosject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2003/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisession:
dev.cdrom.info = Can read MCH:
dev.cdrom.info = Reports media changed:
yoga@MSI:/tmp/test_dir$ dmesg | tail -30
[ 46.188797] Operation canceled @p9io.cpp:258 (AcceptAsync)

[ 46.348173] EXT4-fs (sde): unmounting filesystem 8e6e68d7-2437-4687-834d-6be65d2c8dfa.
[ 125.690990] systemd-journald[43]: Time jumped backwards, rotating.
[ 443.358641] EXT4-fs (sde): mounted filesystem 8e6e68d7-2437-4687-834d-6be65d2c8dfa r/w with ordered data mode. Quota mode: none.
[ 445.179592] misc dxg: dxgk: dxgkio.is.feature_enabled: Ioctl failed: -22
[ 445.194101] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 445.195257] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 445.196527] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 445.198190] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 445.202532] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 445.204426] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 445.206015] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 445.208098] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
[ 459.905108] Exception:
[ 459.905197] Operation canceled @p9io.cpp:258 (AcceptAsync)

[ 460.019613] EXT4-fs (sde): unmounting filesystem 8e6e68d7-2437-4687-834d-6be65d2c8dfa.
[ 461.268096] systemd-journald[43]: Time jumped backwards, rotating.
[ 522.190893] systemd-journald[43]: Time jumped backwards, rotating.
[ 567.953437] mini_init (246): drop_caches: 1
[ 750.033321] systemd-journald[43]: Time jumped backwards, rotating.
[ 1877.985168] systemd-journald[43]: Time jumped backwards, rotating.
[ 2760.752533] systemd-journald[43]: Time jumped backwards, rotating.
[ 2897.305132] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 3004.266273] systemd-journald[43]: Time jumped backwards, rotating.
[ 3047.886305] mini_init (246): drop_caches: 1
[ 4128.464089] systemd-journald[43]: Time jumped backwards, rotating.
[ 4522.541685] systemd-journald[43]: Time jumped backwards, rotating.
[ 4748.311996] systemd-journald[43]: Time jumped backwards, rotating.
yoga@MSI:/tmp/test_dir$ lsmod
Module                  Size Used by
tls                     98304 0
intel_rapl_msr          16384 0
intel_rapl_common       32768 1 intel_rapl_msr
kvm_intel               356352 0
kvm                     970752 1 kvm_intel
```

8. Operating-System Structure

Praktik 8.1: Eksplorasi Struktur Layering

Praktik 8.2: Microkernel vs Monolithic

Praktik 8.3: Program - Virtual File System

```
yoga@MSI:/tmp/test_dir$ lspci
Command 'lspci' not found, but can be installed with:
sudo apt install pciutils
yoga@MSI:/tmp/test_dir$ lsusb
Command 'lsusb' not found, but can be installed with:
sudo apt install usbutils
yoga@MSI:/tmp/test_dir$ cat /proc/cpuinfo

yoga@MSI:/tmp/test_dir$ uname -a
Linux MSI 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 GNU/Linux
yoga@MSI:/tmp/test_dir$ cat /proc/sys/kernel/ostype
linux
yoga@MSI:/tmp/test_dir$ strace ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
yoga@MSI:/tmp/test_dir$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.3 22192 12508 ?        Ss   14:46   0:04 /sbin/init
root         2  0.0  0.0  3072  1792 ?        Sl   14:46   0:00 /init
root         7  0.0  0.0  3088  2060 ?        Sl   14:46   0:00 plan9 --c
root        43  0.0  0.4 66832 18876 ?        S<s  14:46   0:02 /usr/lib/
root        93  0.2  0.1 25184  6348 ?        Ss   14:46   0:10 /usr/lib/
systemd+  108  0.0  0.3 21456 12672 ?        Ss   14:46   0:00 /usr/lib/
systemd+  118  0.0  0.1 91024  7680 ?        Ssl  14:46   0:00 /usr/lib/
root       161  0.0  0.0  4236  2560 ?        Ss   14:46   0:00 /usr/sbin/
message+  162  0.0  0.1  9640  4992 ?        Ss   14:46   0:01 @dbus-dae
root      170  0.0  0.2 17964  8192 ?        Ss   14:46   0:00 /usr/lib/
root      172  0.0  0.3 1756096 12544 ?       Ssl  14:46   0:01 /usr/libe
syslog    189  0.0  0.1 222508  5376 ?       Ssl  14:46   0:00 /usr/sbin/
root     205  0.0  0.5 107012 22400 ?       Ssl  14:46   0:00 /usr/bin/
root     294  0.0  0.0  3076   896 ?        Ss   14:46   0:00 /init
root     295  0.0  0.0  3092  1152 ?        S   14:46   0:04 /init
yoga     296  0.0  0.1  6072  4608 pts/0    Ss   14:46   0:01 -bash
root     297  0.0  0.1  6820  4352 pts/1    Ss   14:46   0:00 /bin/logi
yoga     342  0.0  0.2 203088 11008 ?       Ss   14:46   0:00 /usr/lib/
yoga     343  0.0  0.0 21148  3516 ?        S   14:46   0:00 (sd-pam)
yoga     367  0.0  0.1  6056  5120 pts/1    S+   14:46   0:00 -bash
root     427  0.0  0.0  3116  1792 tty1     Ss+  14:46   0:00 /sbin/age
root     436  0.0  0.0  3160  1920 hvc0     Ss+  14:46   0:00 /sbin/age
polkitd   1014  0.0  0.1 308164  7680 ?       Ssl  15:39   0:00 /usr/lib/
root    1844  0.0  0.0  25188  3696 ?        S   16:13   0:00 (udev-wor
root    1845  0.0  0.0  25188  3696 ?        S   16:13   0:00 (udev-wor
yoga    1846 200  0.1  8280  4096 pts/0    R+   16:13   0:00 ps aux
yoga@MSI:/tmp/test_dir$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.3 22192 12508 ?        Ss   14:46   0:04 /sbin/init
root         2  0.0  0.0  3072  1792 ?        Sl   14:46   0:00 /init
root         7  0.0  0.0  3088  2060 ?        Sl   14:46   0:00 plan9 --c
root        43  0.0  0.4 66832 18876 ?        S<s  14:46   0:02 /usr/lib/
root        93  0.2  0.1 25184  6348 ?        Ss   14:46   0:10 /usr/lib/
systemd+  108  0.0  0.3 21456 12672 ?        Ss   14:46   0:00 /usr/lib/
systemd+  118  0.0  0.1 91024  7680 ?        Ssl  14:46   0:00 /usr/lib/
root       161  0.0  0.0  4236  2560 ?        Ss   14:46   0:00 /usr/sbin/
message+  162  0.0  0.1  9640  4992 ?        Ss   14:46   0:01 @dbus-dae
root      170  0.0  0.2 17964  8192 ?        Ss   14:46   0:00 /usr/lib/
root      172  0.0  0.3 1756096 12544 ?       Ssl  14:46   0:01 /usr/libe
syslog    189  0.0  0.1 222508  5376 ?       Ssl  14:46   0:00 /usr/sbin/
root     205  0.0  0.5 107012 22400 ?       Ssl  14:46   0:00 /usr/bin/
root     294  0.0  0.0  3076   896 ?        Ss   14:46   0:00 /init
root     295  0.0  0.0  3092  1152 ?        S   14:46   0:04 /init
yoga     296  0.0  0.1  6072  4608 pts/0    Ss   14:46   0:01 -bash
root     297  0.0  0.1  6820  4352 pts/1    Ss   14:46   0:00 /bin/logi
yoga     342  0.0  0.2 203088 11008 ?       Ss   14:46   0:00 /usr/lib/
yoga     343  0.0  0.0 21148  3516 ?        S   14:46   0:00 (sd-pam)
yoga     367  0.0  0.1  6056  5120 pts/1    S+   14:46   0:00 -bash
root     427  0.0  0.0  3116  1792 tty1     Ss+  14:46   0:00 /sbin/age
root     436  0.0  0.0  3160  1920 hvc0     Ss+  14:46   0:00 /sbin/age
polkitd   1014  0.0  0.1 308164  7680 ?       Ssl  15:39   0:00 /usr/lib/
yoga    1849  0.0  0.1  8280  4096 pts/0    R+   16:13   0:00 ps aux
yoga@MSI:/tmp/test_dir$ top
top - 16:14:50 up 1:28, 1 user, load average: 0.00, 0.00, 0.00

yoga@MSI:/tmp/test_dir$ which bash
/usr/bin/bash
yoga@MSI:/tmp/test_dir$ which gcc
yoga@MSI:/tmp/test_dir$ ls -lh /boot/vmlinuz-$(uname -r)
ls: cannot access '/boot/vmlinuz-6.6.87.2-microsoft-standard-WSL2': No such file or directory
yoga@MSI:/tmp/test_dir$ cat /boot/config-$(uname -r) | grep "=y" | wc -l # Built-in
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
yoga@MSI:/tmp/test_dir$ cat /boot/config-$(uname -r) | grep "=m" | wc -l # Modules
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
yoga@MSI:/tmp/test_dir$ lsmod | wc -l
19
yoga@MSI:/tmp/test_dir$ nano procfs_read.c
yoga@MSI:/tmp/test_dir$ gcc -o procfs_read procfs_read.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ ./procfs_read
```

9. Building and Booting an Operating System

Praktik 9.1: Analisis Boot Process

Praktik 9.2: Kernel Parameters

Praktik 9.3: Simple Bootloader (Konsep)

```
yoga@MSI:/tmp/test_dir$ dmesg | less

[1]+  Stopped                  dmesg | less
yoga@MSI:/tmp/test_dir$ ^C
yoga@MSI:/tmp/test_dir$ systemd-analyze
Startup finished in 2.980s (userspace).
graphical.target reached after 2.916s in userspace.
yoga@MSI:/tmp/test_dir$ systemd-analyze blame
1min 37.465s motd-news.service
1.701s landscape-client.service
572ms snapd.seeded.service
544ms wsl-pro.service
427ms snapd.service
398ms dev-sdd.device
380ms console-getty.service
274ms systemd-resolved.service
272ms user@1000.service
246ms systemd-tmpfiles-clean.service
231ms snapd.socket
213ms my_daemon.service
178ms systemd-timesyncd.service
177ms rsyslog.service
124ms systemd-udev-trigger.service
122ms systemd-logind.service
110ms systemd-udevd.service
93ms polkit.service
76ms keyboard-setup.service
74ms dbus.service
68ms e2scrub_reap.service
66ms systemd-journald.service
51ms systemd-journal-flush.service
47ms dev-hugepages.mount
46ms dev-mqueue.mount
45ms sys-kernel-debug.mount
44ms sys-kernel-tracing.mount
39ms user-runtime-dir@1000.service
36ms systemd-tmpfiles-setup-dev-early.service
35ms systemd-sysctl.service
33ms kmod-static-nodes.service
33ms modprobe@configfs.service
33ms modprobe@drm.service
32ms modprobe@efi_pstore.service
32ms modprobe@dm_mod.service
32ms systemd-tmpfiles-setup.service
32ms modprobe@fuse.service
30ms modprobe@loop.service
29ms systemd-update-utmp.service
27ms systemd-modules-load.service

Lines 1-40
[2]+  Stopped                  systemd-analyze blame
yoga@MSI:/tmp/test_dir$ [200~systemd-analyze critical-chain-
[200~systemd-analyze: command not found
yoga@MSI:/tmp/test_dir$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
yoga@MSI:/tmp/test_dir$ lsinitramfs /boot/initrd.img-$(uname -r) | head -20
Command 'lsinitramfs' not found, but can be installed with:
sudo apt install initramfs-tools-core
yoga@MSI:/tmp/test_dir$ cat /proc/cmdline
initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
```

10. Operating-System Debugging

Praktik 10.1: Using strace

Praktik 10.2: Using ltrace

Praktik 10.3: Using GDB

Praktik 10.4: Kernel Debugging dengan dmesg

Praktik 10.5: Memory Leak Detection dengan Valgrind

```
yoga@MSI:/tmp/test_dir$ strace ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
yoga@MSI:/tmp/test_dir$ ^C
yoga@MSI:/tmp/test_dir$ strace ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
yoga@MSI:/tmp/test_dir$ strace -v ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
yoga@MSI:/tmp/test_dir$ strace -c ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
yoga@MSI:/tmp/test_dir$ strace -o trace.txt ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
yoga@MSI:/tmp/test_dir$ cat trace.txt
cat: trace.txt: No such file or directory
yoga@MSI:/tmp/test_dir$ ltrace ls
Command 'ltrace' not found, but can be installed with:
sudo apt install ltrace
yoga@MSI:/tmp/test_dir$ ltrace -c ls
Command 'ltrace' not found, but can be installed with:
sudo apt install ltrace
yoga@MSI:/tmp/test_dir$ nano buggy.c
yoga@MSI:/tmp/test_dir$ gcc -g -o buggy buggy.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ gdb ./buggy
Command 'gdb' not found, but can be installed with:
sudo apt install gdb
yoga@MSI:/tmp/test_dir$ sudo apt install gdb
[sudo] password for yoga:
Sorry, try again.
[sudo] password for yoga:
E: Type '-' is not known on line 1 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo apt install gdb
E: Type '-' is not known on line 1 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo apt install gdb
E: Type '-' is not known on line 1 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo apt install gdb
E: Type '-' is not known on line 1 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo nano /etc/apt/sources.list
yoga@MSI:/tmp/test_dir$ dmesg
```

```

yoga@MSI:/tmp/test_dir$ sudo apt install valgrind
E: Type '(gdb)' is not known on line 1 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ nano /etc/apt/sources.list
yoga@MSI:/tmp/test_dir$ nano sudo apt install valgrind
yoga@MSI:/tmp/test_dir$ sudo apt install valgrind
E: Type '(gdb)' is not known on line 1 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ gdb ./buggy
Command 'gdb' not found, but can be installed with:
sudo apt install gdb
yoga@MSI:/tmp/test_dir$ gcc -g -o buggy buggy.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ nano buggy.c
yoga@MSI:/tmp/test_dir$ gcc -g -o buggy buggy.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ sudo apt install gcc
E: Type '(gdb)' is not known on line 1 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ nano buggy.c
yoga@MSI:/tmp/test_dir$ gdb ./buggy
Command 'gdb' not found, but can be installed with:
sudo apt install gdb
yoga@MSI:/tmp/test_dir$ gdb ./buggy
Command 'gdb' not found, but can be installed with:
sudo apt install gdb
yoga@MSI:/tmp/test_dir$ nano buggy.c
yoga@MSI:/tmp/test_dir$ sudo apt install valgrind
E: Type '(gdb)' is not known on line 1 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo apt install linux-tools-common linux-tools-generic
E: Type '(gdb)' is not known on line 1 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo perf trace ls
sudo: perf: command not found
yoga@MSI:/tmp/test_dir$ sudo perf top
sudo: perf: command not found
yoga@MSI:/tmp/test_dir$ sudo apt install valgrind
E: Type '(gdb)' is not known on line 1 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo nano /etc/apt/sources.list
yoga@MSI:/tmp/test_dir$ sudo nano /etc/apt/sources.list
yoga@MSI:/tmp/test_dir$ sudo apt update
E: Type '(gdb)' is not known on line 4 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo nano /etc/apt/sources.list
yoga@MSI:/tmp/test_dir$ sudo apt update
sudo apt install valgrind
E: Type '(gdb)' is not known on line 3 in source list /etc/apt/sources.list
E: The list of sources could not be read.
E: Type '(gdb)' is not known on line 3 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo apt update
E: Type '(gdb)' is not known on line 3 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo apt install valgrind
E: Type '(gdb)' is not known on line 3 in source list /etc/apt/sources.list
E: The list of sources could not be read.
yoga@MSI:/tmp/test_dir$ sudo nano /etc/apt/sources.list
yoga@MSI:/tmp/test_dir$ sudo apt update

```

```

yoga@MSI:/tmp/test_dir$ cat > memleak.c << 'EOF'
> ^C
yoga@MSI:/tmp/test_dir$ cat > memleak.c << 'EOF'
> ^C
yoga@MSI:/tmp/test_dir$ cat > memleak.c << 'EOF'
> int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
    return 0;
}
EOF
gcc -g -o memleak memleak.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ cat > memleak.c << 'EOF'
> int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
    return 0;
}
>
>
>
>
> ^C
yoga@MSI:/tmp/test_dir$ cat > memleak.c << 'EOF'
> #include <stdlib.h>
int main() {
    int *ptr = (int*)malloc(100 * sizeof(int));
    // Forgot to free(ptr)
    return 0;
}
EOF
yoga@MSI:/tmp/test_dir$ gcc -g -o memleak memleak.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
yoga@MSI:/tmp/test_dir$ sudo apt isntall gcc
E: Invalid operation isntall
yoga@MSI:/tmp/test_dir$ sudo apt install gcc
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  cpp cpp-13 cpp-13-x86-64-linux-gnu cpp-x86-64-linux-gnu gcc-13
  gcc-13-base gcc-13-x86-64-linux-gnu gcc-x86-64-linux-gnu libaom3
  libasan8 libatomic1 libc-dev-bin libc-devtools libc6-dev libcc1-0
  libcrypt-dev libde265-0 libgcc-13-dev libgd3 libgomp1
  libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265
  libheif1 libhwasan0 libisl23 libitm1 liblsan0 libmpc3 libquadmath0
  libtsan2 libubsan1 libxpm4 linux-libc-dev manpages-dev rpcsvc-proto

```



```

yoga@MSI:/tmp/test_dir$ gcc -g -o memleak memleak.c
yoga@MSI:/tmp/test_dir$ valgrind --leak-check=full ./memleak
==4434== Memcheck, a memory error detector
==4434== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==4434== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==4434== Command: ./memleak
==4434==
==4434==
==4434== HEAP SUMMARY:
==4434==   in use at exit: 400 bytes in 1 blocks
==4434==   total heap usage: 1 allocs, 0 frees, 400 bytes allocated
==4434==
==4434== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1
==4434==   at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==4434==   by 0x10915E: main (memleak.c:3)
==4434==
==4434== LEAK SUMMARY:
==4434==   definitely lost: 400 bytes in 1 blocks
==4434==   indirectly lost: 0 bytes in 0 blocks
==4434==   possibly lost: 0 bytes in 0 blocks
==4434==   still reachable: 0 bytes in 0 blocks
==4434==   suppressed: 0 bytes in 0 blocks
==4434==
==4434== For lists of detected and suppressed errors, rerun with: -s
==4434== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
yoga@MSI:/tmp/test_dir$ sudo apt install linux-tools-common linux-tools-generic
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  hwdata libnumal libpci3 libtraceevent1 linux-tools-6.8.0-85
  linux-tools-6.8.0-85-generic
The following NEW packages will be installed:
  hwdata libnumal libpci3 libtraceevent1 linux-tools-6.8.0-85
  linux-tools-6.8.0-85-generic linux-tools-common linux-tools-generic
0 upgraded, 8 newly installed, 0 to remove and 50 not upgraded.
Need to get 6340 kB of archives.
After this operation, 21.0 MB of additional disk space will be used.
Do you want to continue? [Y/n] sudo perf trace ls
Abort.
yoga@MSI:/tmp/test_dir$ sudo perf top
sudo: perf: command not found
yoga@MSI:/tmp/test_dir$ sudo perf trace ls
sudo: perf: command not found
yoga@MSI:/tmp/test_dir$ sudo perf top
sudo: perf: command not found
yoga@MSI:/tmp/test_dir$ |

```


